Social Networks and their Analysis

AN INTRODUCTION TO THE FIELD

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Social Networks and their Analysis

An Introduction to the field

- **Motivation**
  - **Link Analysis**
  - *SF-networks*
- *Background*
  - *Application Examples*

Basic Concepts

- Network Representation
- Tie Strength
- Key Players
- Cohesion

Motivation: Link Analysis

Objectives:

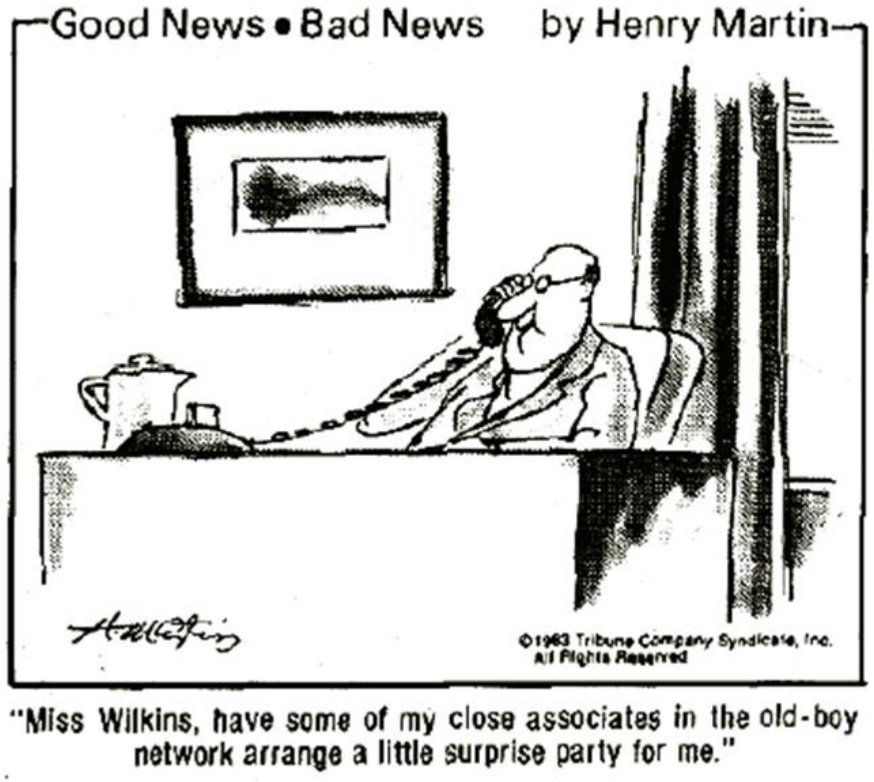
- ***find relationships among the data***
- ***visualize the links and relationships***

Application areas:

- telecommunications
- criminalistics and law – groups of criminals are mutually interlinked, an analysis of these relationships can help uncover them
- marketing – relationships among customers

Motivation: Link Analysis

High schools as networks

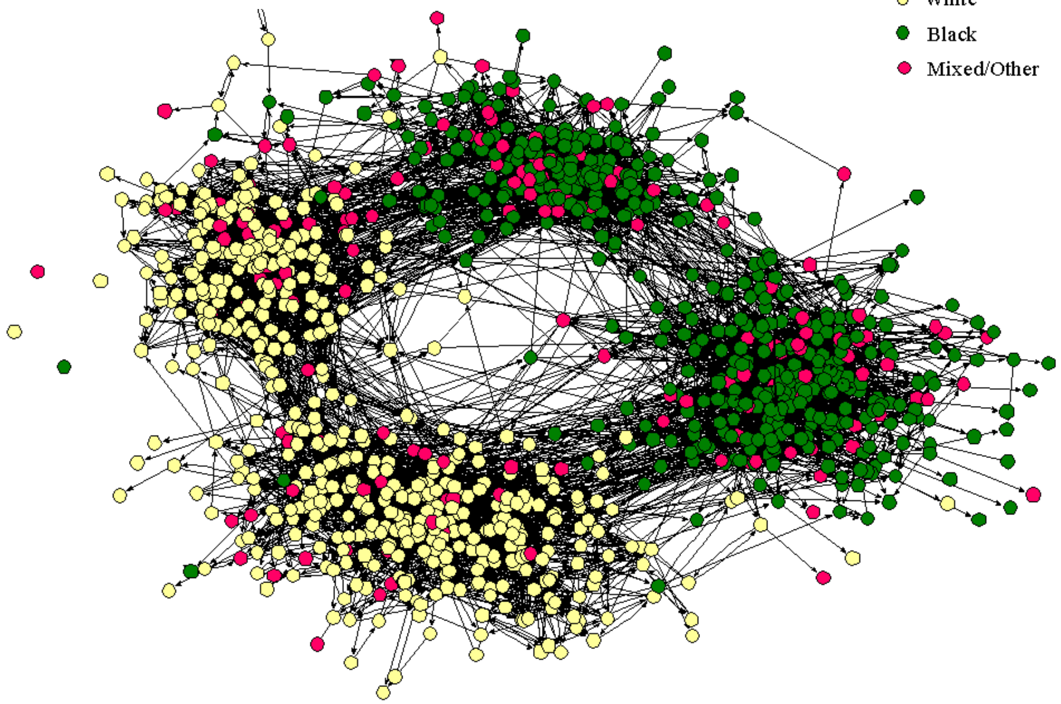


Source: Linton Freeman "See you in the funny pages"
Connections, 23, 2000, 32-42.

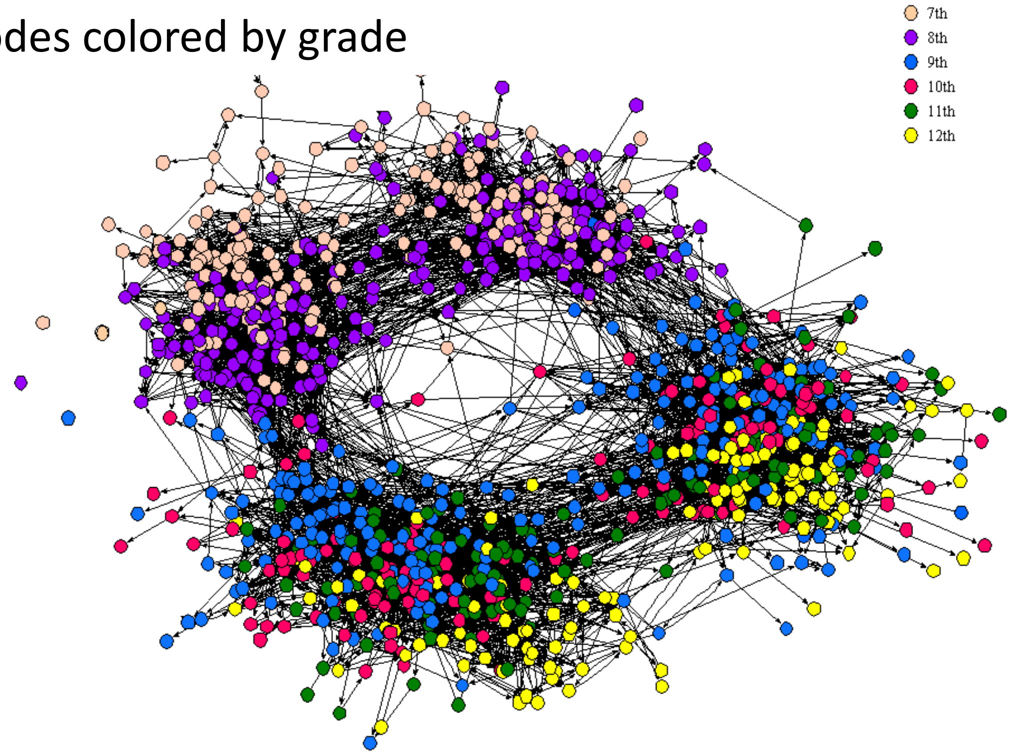
Motivation: Link Analysis

Social (friendship) structure of the „Countryside“ school district

Nodes colored by race



Nodes colored by grade



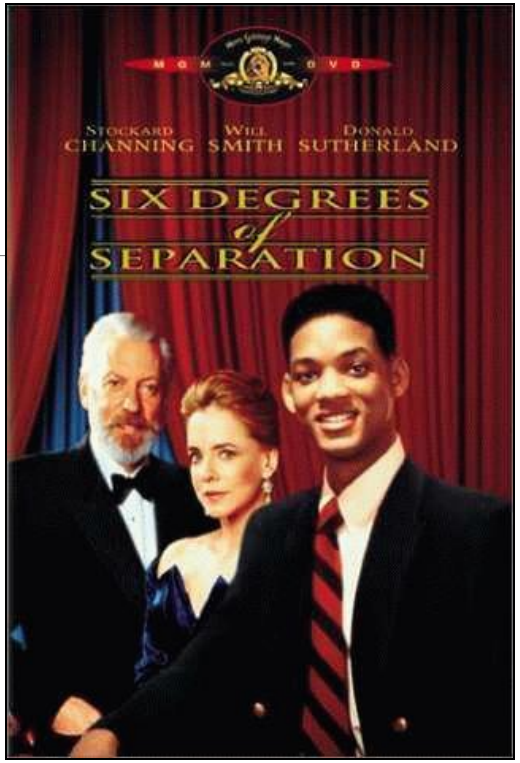
Motivation: Link Analysis


Stanley Milgram (1933-1984)

Six Degrees of Separation, (Milgram, 1967)

- an experiment on the small-worlds phenomenon
- random people from Nebraska were asked to send a letter (via intermediaries) to a stock-broker in Boston
  - could only send to someone whom they knew on a first-name basis
- among the letters that found the target, the average number of links was six
  - However, not many of the letters in the original experiment arrived
  - the experiment presaged some later discoveries (e.g., hubs, tendency to go by geography, profession, etc.)

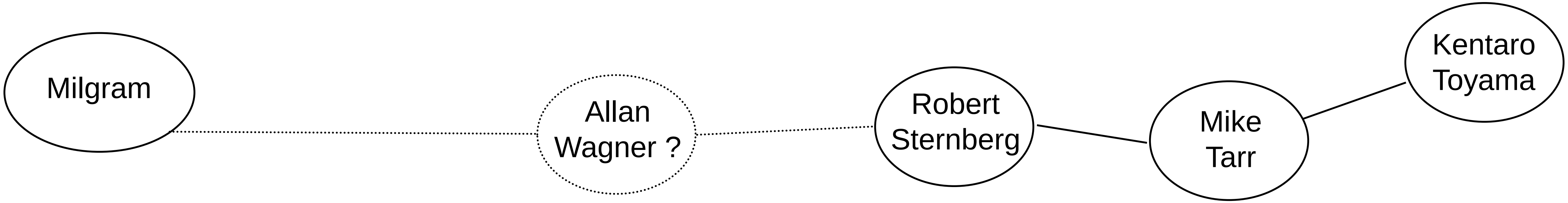
Motivation: Link Analysis



Six Degrees of Separation, (Milgram, 1967)

John Guare wrote a play called *Six Degrees of Separation,* based on this concept:

"Everybody on this planet is separated by only six other people. Six degrees of separation. Between us and everybody else on this planet. The president of the United States. A gondolier in Venice… It's not just the big names. It's anyone. A native in a rain forest. A Tierra del Fuegan. An Eskimo. I am bound to everyone on this planet by a trail of six people…"



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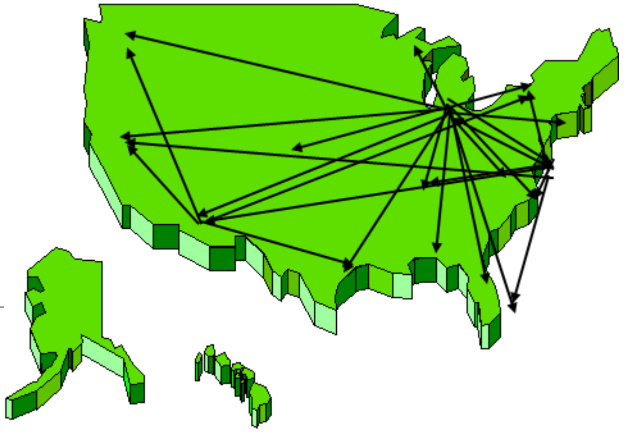
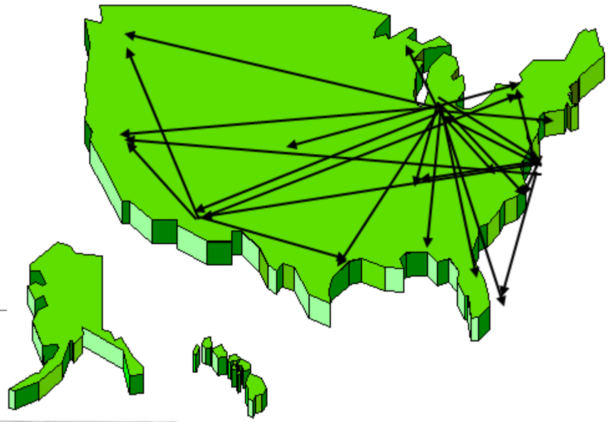
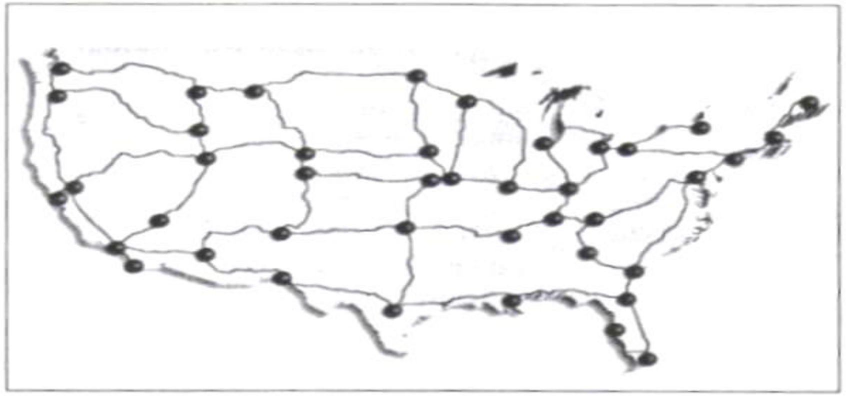
Motivation: SF-Networks (Scale-Free Networks)

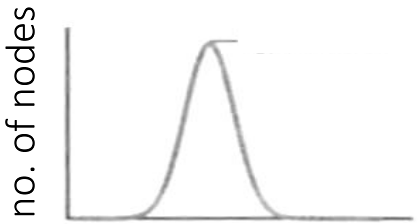


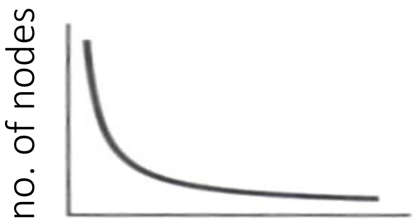
- Some nodes have extremely many connections (edges) to other nodes - **hubs**
- Most nodes have just a few connections to other nodes
- Resilience to a random disturbance
- Vulnerability to coordinated attacks
- **New application areas:**
  - protection against (computer) viruses spread through the internet
  - medicine (vaccinations)
  - business (marketing)

SF-Networks









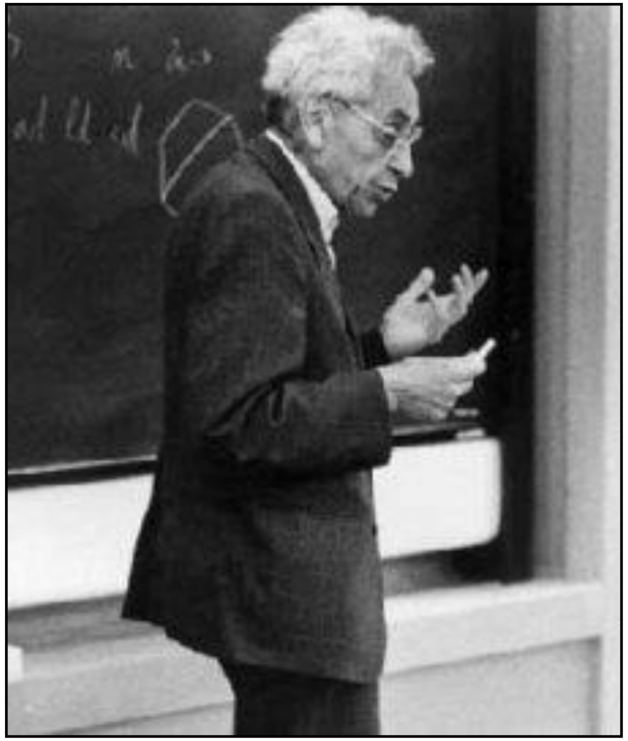
Adapted from "A. L. Barabasi and E. Bonabeau: *Scale-Free Networks*, Scientific American, May 2003"

Examples of SF-Networks

Scientific collaboration - researchers co-authoring papers

- Paul Erdős wrote 1500+ papers with 507 co-authors
- **Erdős number:**
  - ~ no. of links required to connect scholars to Erdős, via co-authorship of papers
  - Jerry Grossman's (Oakland Univ.) website allows mathematicians to compute their Erdos numbers:
    https://sites.google.com/oakland.edu/grossman/home/the-erdoes-number-project
    https://mathscinet.ams.org/mathscinet/freeTools.html?version=2

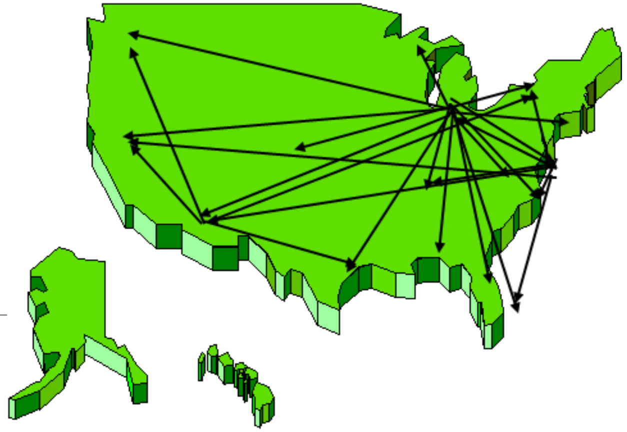
Paul Erdős (1913-1996)



connecting path lengths, among mathematicians only:

- average Erdős number is *4.65,* maximum is *13*

Examples of SF-Networks



Social networks

- Scientific collaboration – researchers co-authoring papers
- Hollywood – actors playing roles in the same movie

Biological networks

- cell metabolism

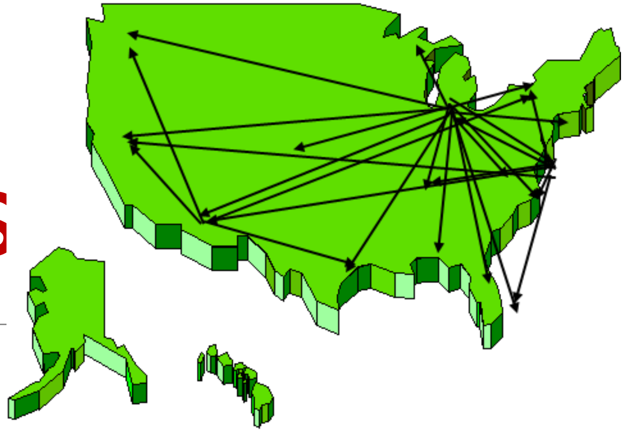
  \- molecules participating in energy production, in the same biological reaction
- protein regulatory networks

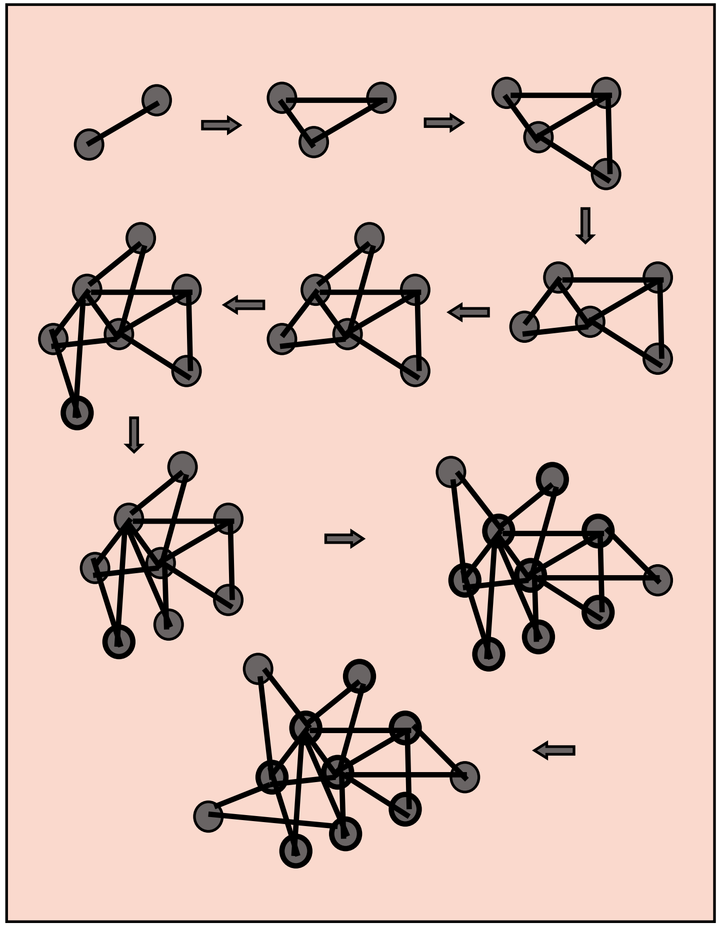
  \- proteins that control cell activity, interaction between proteins

Socio-technical networks

- Internet – routers, optical and other connections
- World Wide Web – web pages and URL (~ Uniform Resource Locator)

SF-Networks: Basic Characteristics





Two main mechanisms:

- **growth**
- **preferential attachment**

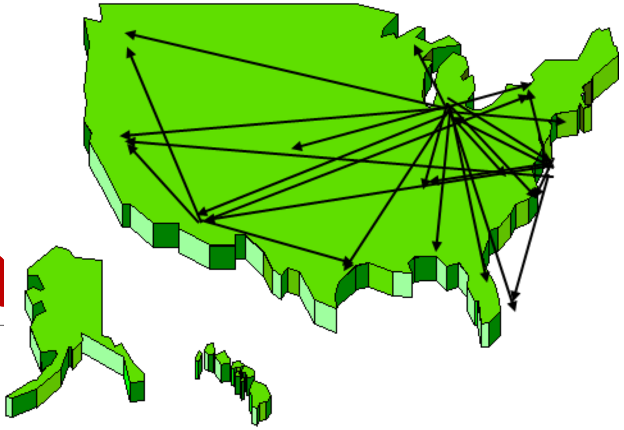
"The rich get richer" (hubs):

- new nodes prefer to attach themselves to nodes already equipped with more links
- During the course of time, "popular locations" form more connections than their neighbors who have less links

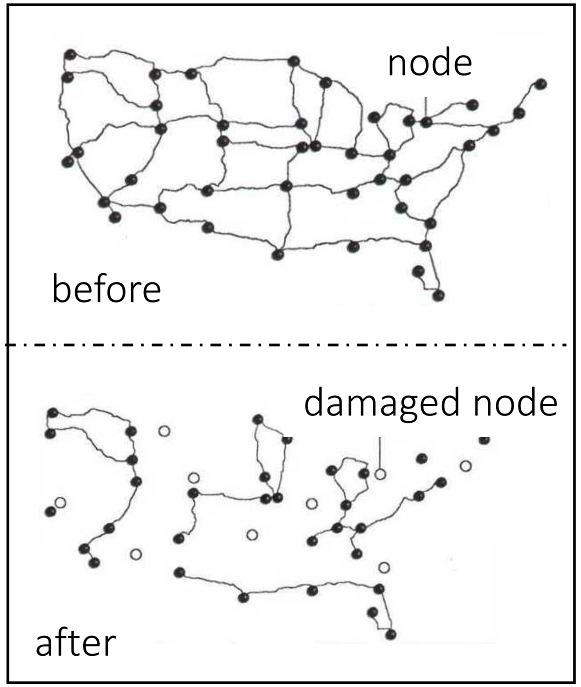
Reliability

- ***random failures*** (80% *of randomly chosen nodes* can fail without leading to the fragmentation of the cluster)
- ***coordinated attacks*** (the elimination of 5-15% *of all the hubs* can cause the entire system to collapse)

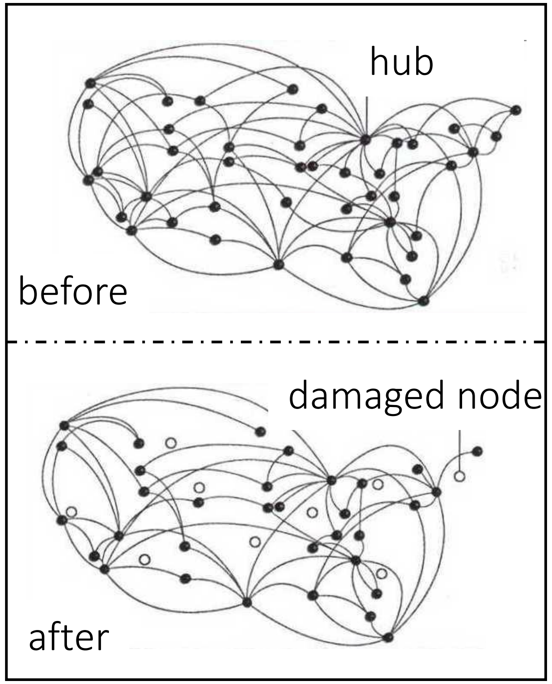
SF-Networks:
an Example Resembling City Connection



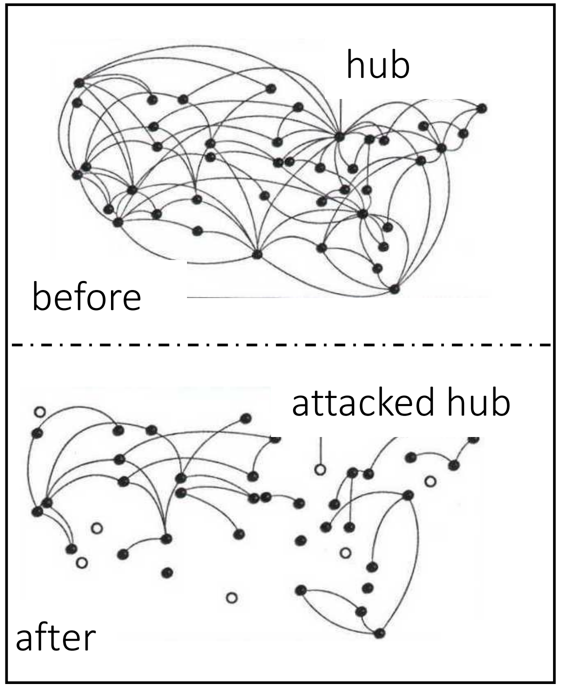
A random network: failure of a random node



A SF- network: failure of a random node

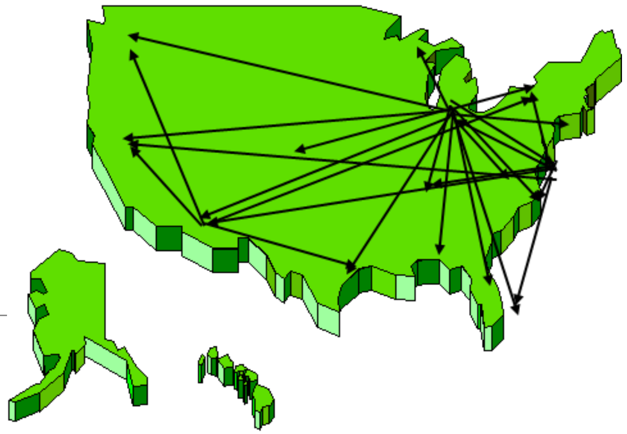


A SF-network: co-ordinated attack of hubs



Adapted from "A. L. Barabasi and E. Bonabeau: *Scale-Free Networks*, Scientific American, May 2003"

Applications of SF-Networks



Computing

- networks with a Scale-Free-architecture

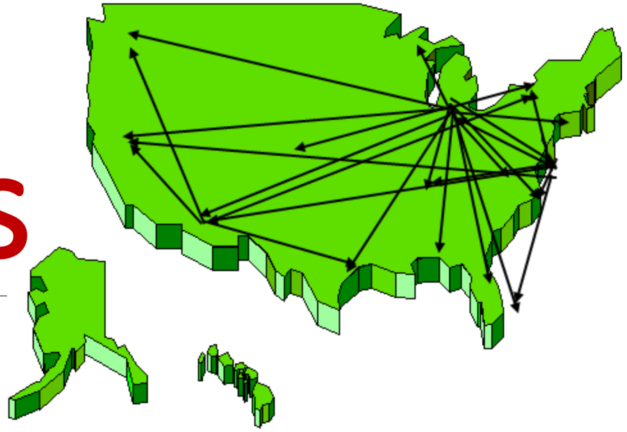
Medicine

- vaccination campaigns and new drugs

Business

- cascaded financial crashes
- marketing (word of mouth, influencer marketing, social media and search engine marketing, …)

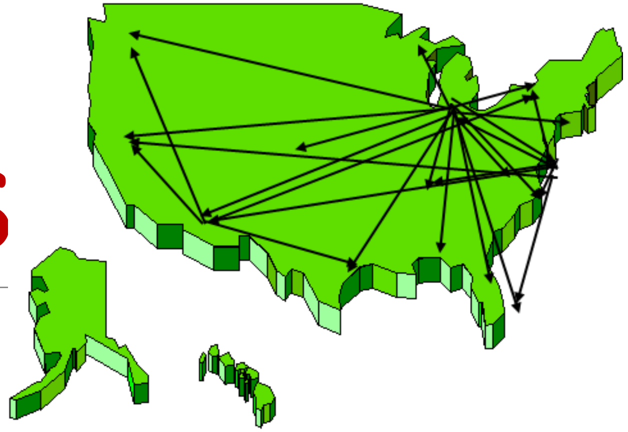
Application of SF-Networks



Computing

- Computer networks with a SF-architecture (e.g., WWW)
  - Extremely resistant to random failures
  - Very vulnerable to coordinated attacks or sabotage
- Eradication of internet viruses is essentially impossible

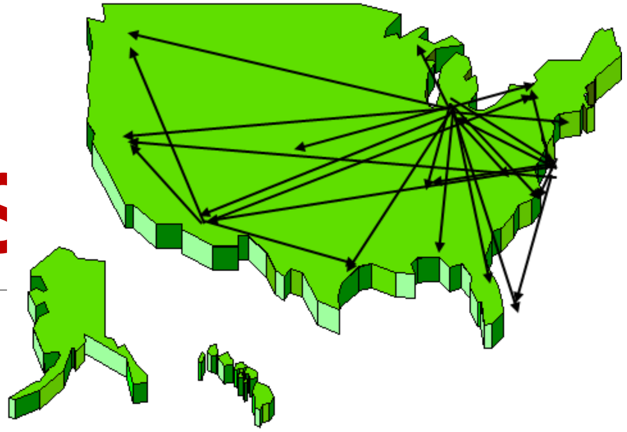
Application of SF-Networks



Medicine

- Vaccination campaigns focused on hubs
  - People with many contacts and social connections
  - A difficult identification of such individuals
- New drugs targeting the key molecules (hubs) involved in the respective disease
- minimize the side effects of the drugs by using the found link networks inside the cells

Application of SF-Networks



Business

- **Financial failures**
  - understanding of mutual links between companies, industry and economy
  - monitoring and elimination of cascade financial crashes
- **Marketing**
  - explore the spread of contagion in SF-networks
  - a more efficient advertising of new products

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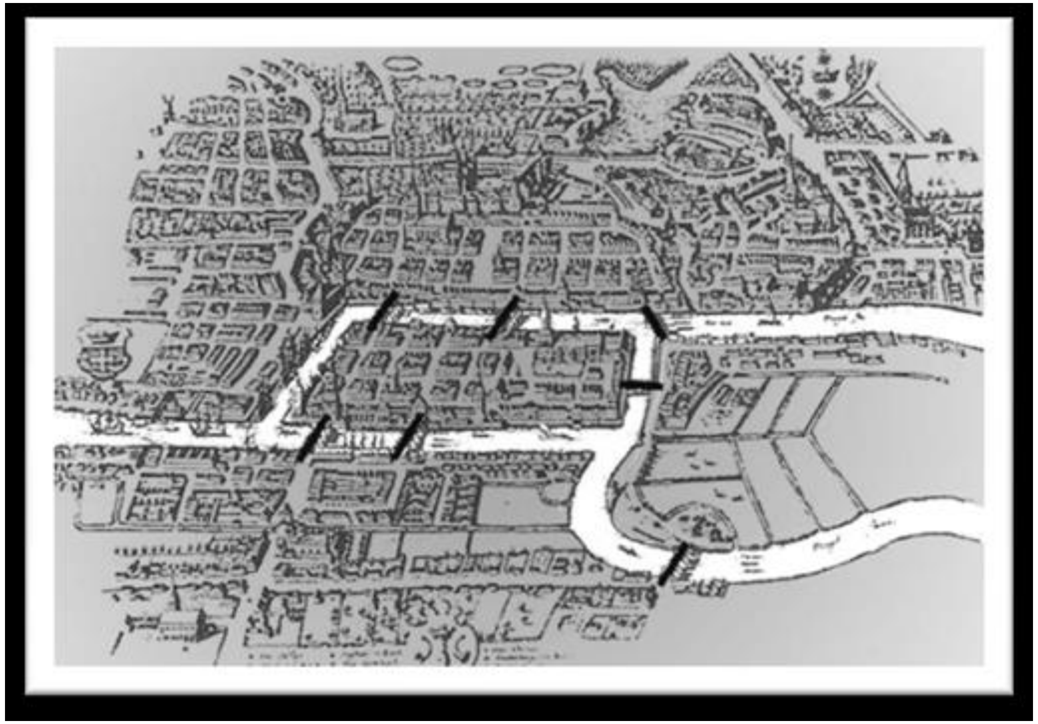
Background: Social Network Analysis

Social Network Analysis (SNA)

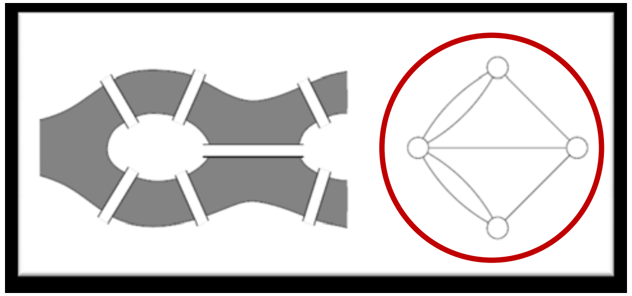
- **interdisciplinary field** with origin in social science, network analysis and graph theory
  - **social science** – questions and problems to explain social behavior through an analysis of societies
  - **network analysis** - formulation and solution of problems that have a network structure
  - **graph theory** – a mathematical field that provides a set of abstract concepts and methods to represent and analyze graphs.
- **focused on relationships** established between social entities (individuals, groups, institutions, etc.) rather than on the social entities themselves (i.e., individuals or their attributes)
  - SNA methods include **visualization** and **analytical tools**.

Background: Social Network Analysis



An early example of network analysis:

- city of Königsberg (now Kaliningrad).
- Leonard Euler proved that there is no path that crosses each of the city's bridges only once (Newman et al, 2006).



Background: Social Science

- Studying society from a network perspective considers **individuals** as **embedded in a network of relations** and seeks **explanations** for social behavior **in the structure of these networks** rather than in the individuals alone.
- **widespread availability of data,** as well as advances in computing and methodology, make it easier to apply SNA

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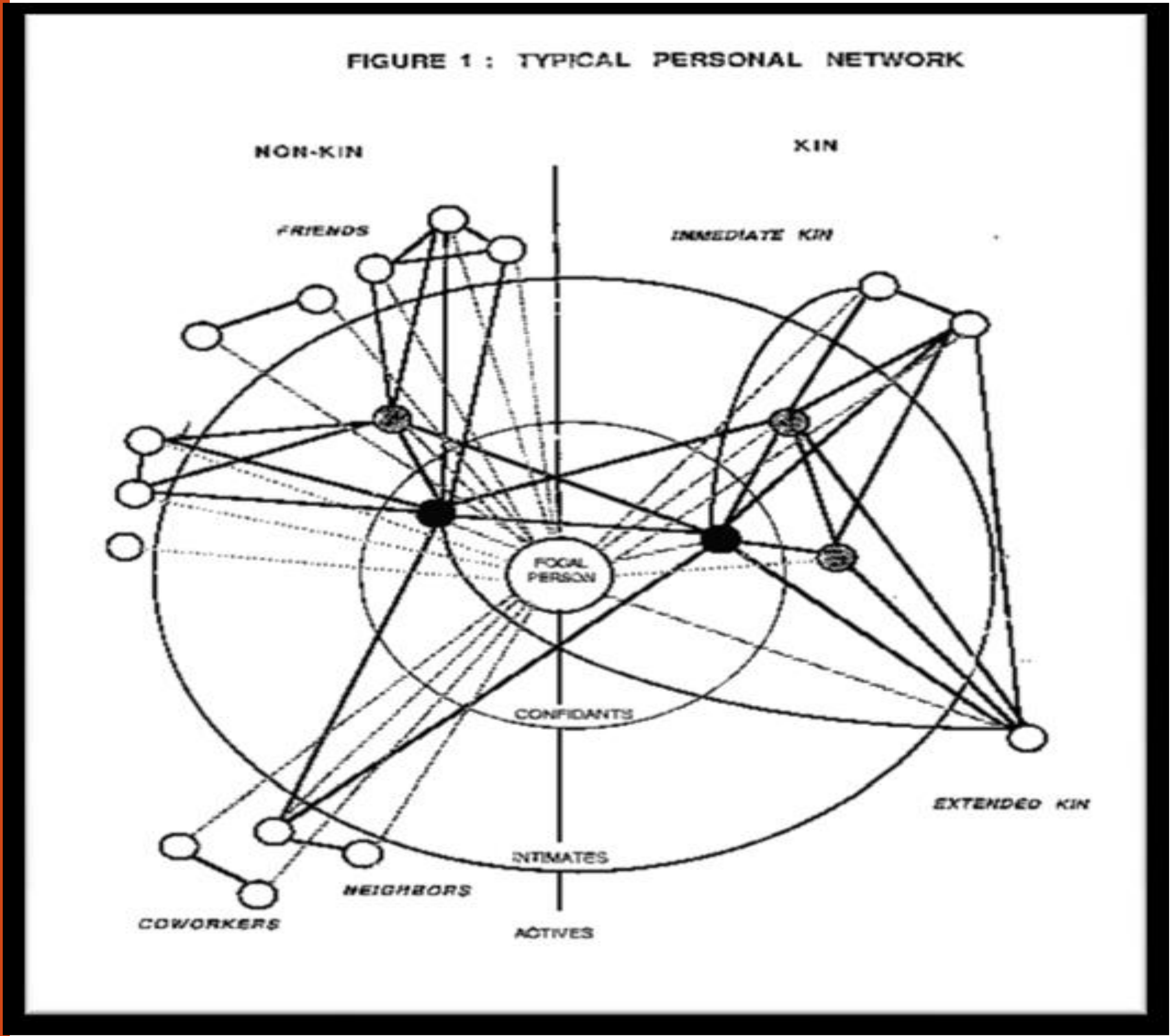
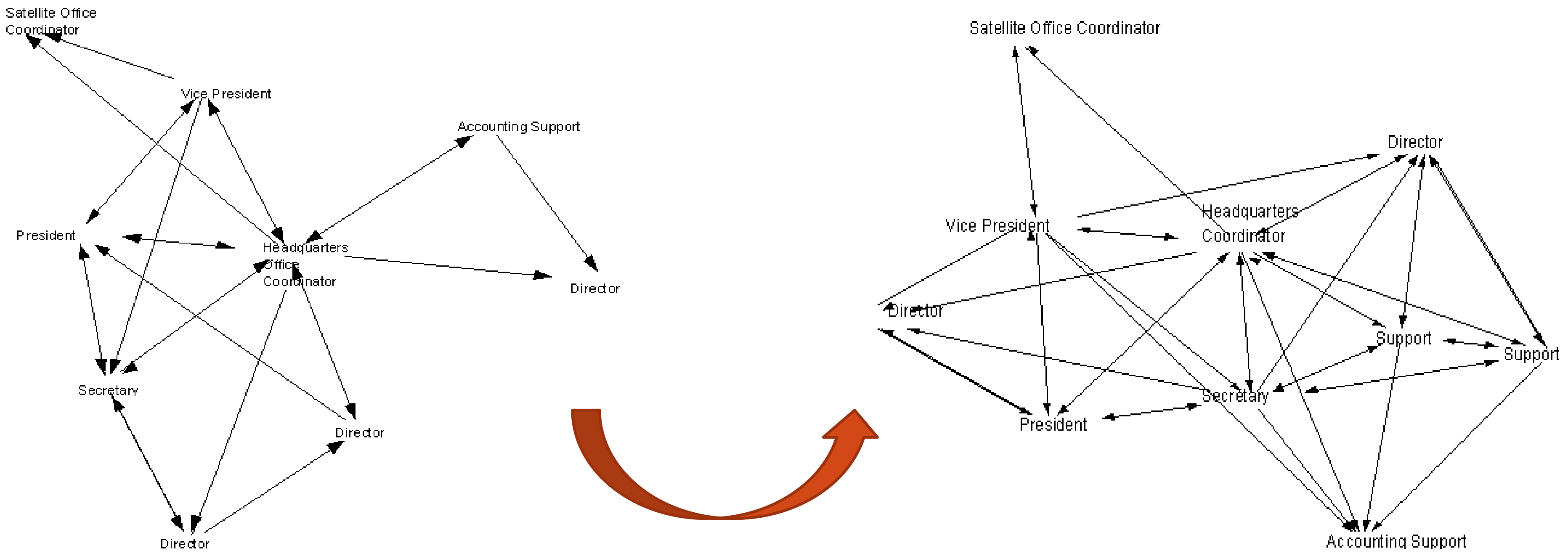
An early depiction of an ‘ego’ network:

- a personal network.
- varying tie strengths depicted via concentric circles (Wellman, 1998)

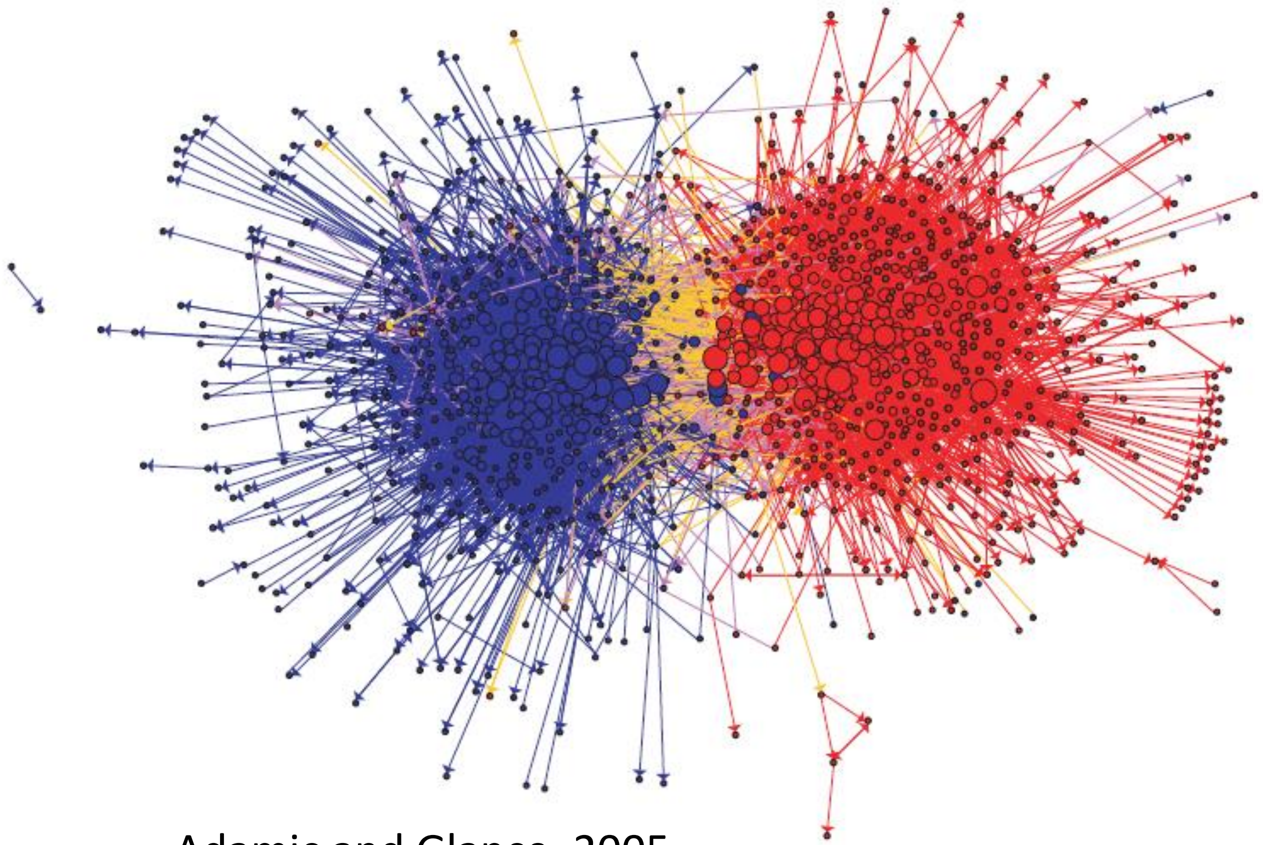


More Examples from Social Science

These visualizations depict the flow of communications in an organization before and after the introduction of a content management system (Gartonet al, 1997)



More Examples from Social Science



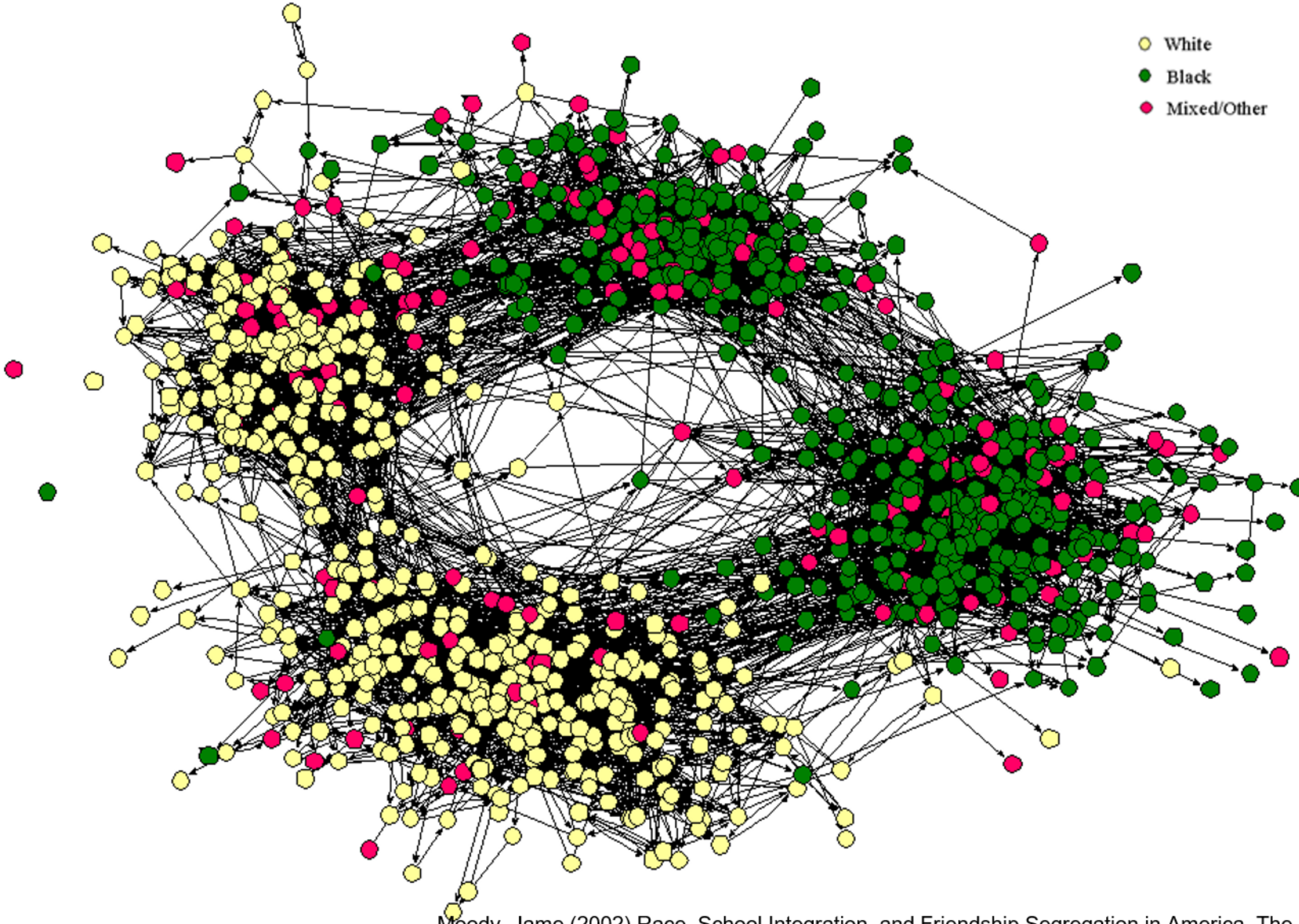
Adamic and Glance, 2005

Visualization of US bloggers:

- form two distinct clusters
- tend to link predominantly to blogs supporting the same party

Race & school friendships

Black & white friendship nominations – 4 clusters are white/black



Moody, Jame (2002) Race, School Integration, and Friendship Segregation in America. The American journal of sociology [0002-9602] Moody yr:2002 vol:107 iss:3 pg:679

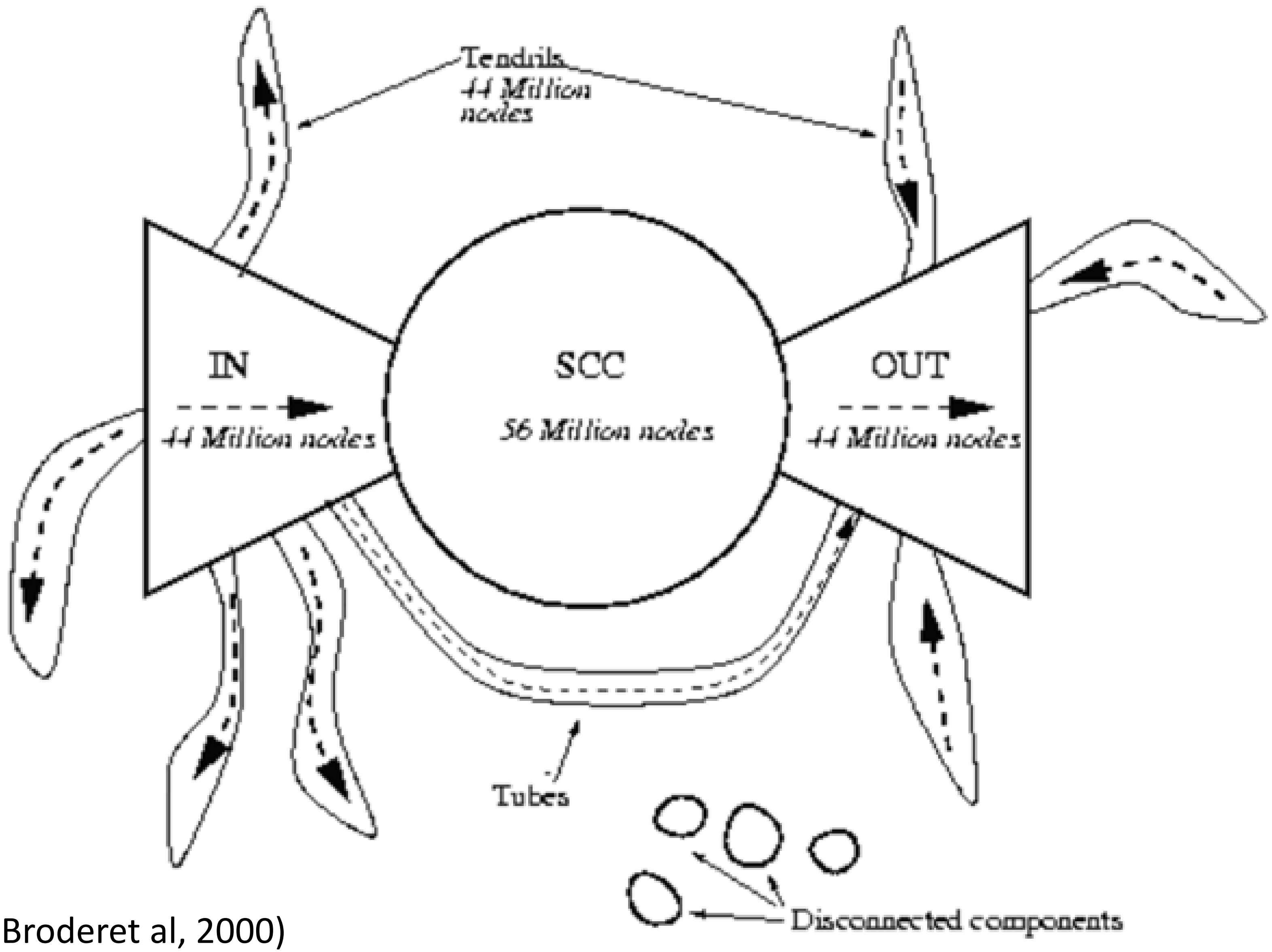
Background: Other Domains for SNA

(Social) Network Analysis is predominantly used to study human-generated structures, but it has found applications in many domains beyond:

- Computer science – analysis of webpages, internet traffic, information dissemination, etc.
- Life sciences – a study of food chains in different ecosystems
- Mathematics and (theoretical) physics – analysis of networks from relevant domains

Links between web pages

- a large-scale human-generated structure
- the Web consists of a core of densely inter-linked pages
- most other web pages either linked to or lin-ked from the core.



(Broderet al, 2000)

Practical Applications

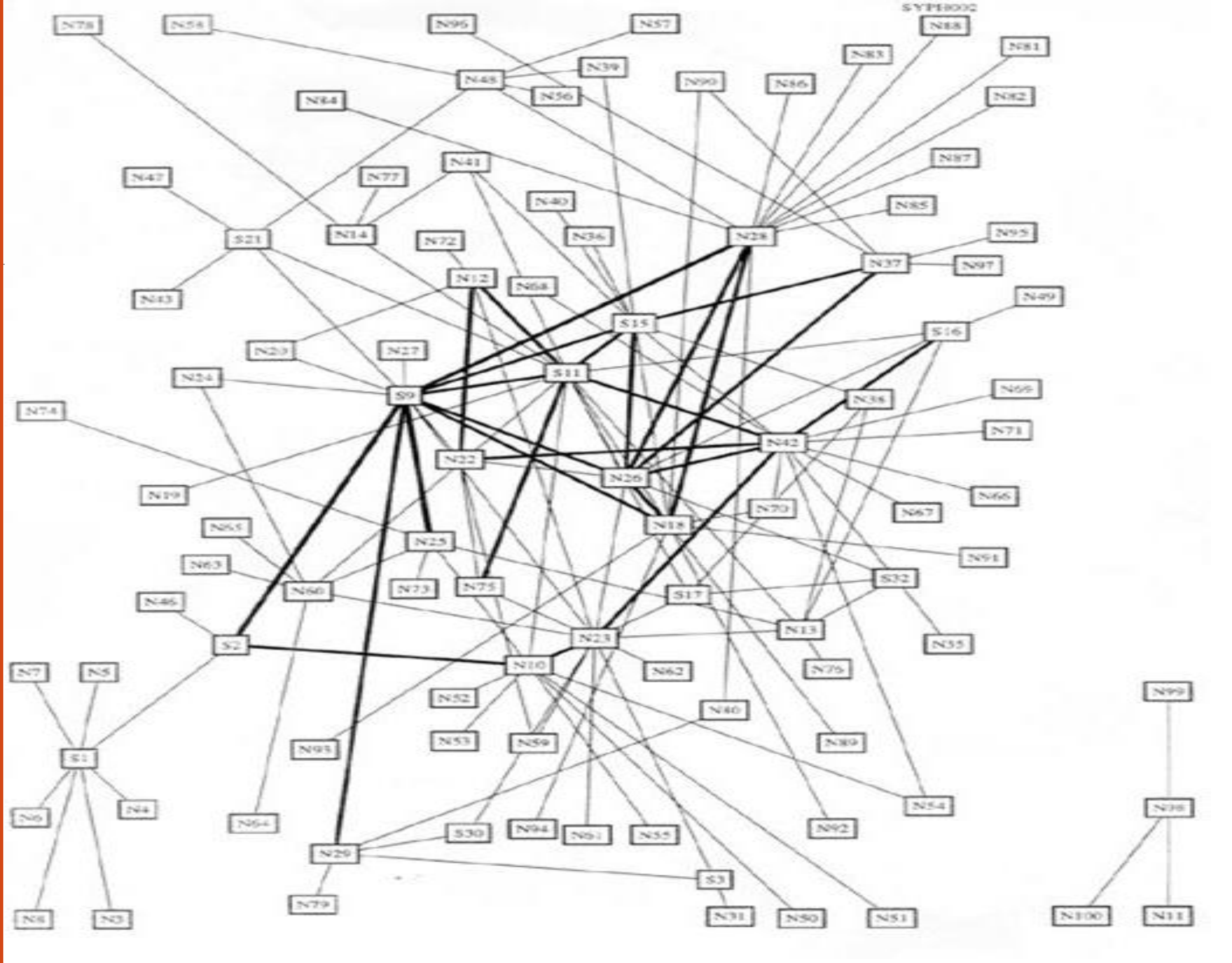
- Businesses:
  - analyze and improve communication flow in an organization, with partners or customers
  - marketing campaigns based on social networking
- Law enforcement agencies (and the army):
  - identify criminal and terrorist networks from traces of collected communication
  - identify key players in criminal or terrorist networks
- Social Network Sites (like Facebook):
  - identify and recommend potential friends based on friends-of-friends

Practical Applications

- Civil society organizations:
  - uncover conflicts of interest in hidden connections between government bodies, lobbies and businesses
- Network operators (telephony, cable, mobile):
  - optimize the structure and capacity of communication networks
- Medicine:
  - Contagious diseases and their prevention
  - Sexually transmitted diseases (STD) – 1996 syphilis outbreak in Rockdale County (Atlanta metropolitan area, Georgia)

Syphilis outbreak at a high school in Rockdale

- affluent suburb of Atlanta, Georgia
- a group of young white girls about 16 years old
- a group of white boys aged 17 to 21
- a group of African-American boys 17 to 21
- Syphilis was diagnosed in 6 white females, 2 white males and 2 African-American males



Why and When to Use SNA

- study a social network (offline or online) or improve its effectiveness
- visualize data so as to uncover patterns in relationships or interactions
- follow the paths that information flows in social networks
- in quantitative research (yet for qualitative research, a network perspective is also valuable):
  - the range of actions and opportunities afforded to individuals are often a function of their positions in social networks; uncovering these positions (say as fathers, mothers, teachers, workers) can yield interesting and sometimes surprising results
  - quantitative analysis of a social network can help identify different types of actors in the network (key players)
- analyze social media in general, identify the causes for dysfunctional communities or networks, and promote social cohesion and growth in an online community